

**I BUILT A BRUTE FORCE
DETECTION LAB.**

**The real attackers
showed up first.**

Swipe to see how →

First, I looked at my server the way an attacker would. nmap pulled the SSH version in half a second.

```
PS C:\Users\calip> nmap -sV -Pn -p 22 52.246.249.89
Starting Nmap 7.99 ( https://nmap.org ) at 2026-05-26 14:31 -0700
Nmap scan report for 52.246.249.89
Host is up (0.037s latency).

PORT      STATE SERVICE VERSION
22/tcp    open  ssh      OpenSSH 9.6p1 Ubuntu 3ubuntu13.16 (Ubuntu Linux; protocol 2.0)
Service Info: OS: Linux; CPE: cpe:/o:linux:linux_kernel
```

**While I was watching the auth log,
real attackers started showing up.
None of them were me.**

```
2026-05-26T21:37:32.205655+00:00 az900-vm1 sshd[11974]: Failed password for invalid user ubuntu from 35.188.112.111 port 45270 ssh2
2026-05-26T21:37:32.318091+00:00 az900-vm1 sshd[11974]: Received disconnect from 35.188.112.111 port 45270:11: Bye Bye [preauth]
2026-05-26T21:37:32.318178+00:00 az900-vm1 sshd[11974]: Disconnected from invalid user ubuntu 35.188.112.111 port 45270 [preauth]
2026-05-26T21:37:37.018064+00:00 az900-vm1 sshd[11978]: Invalid user ubuntu from 178.27.90.142 port 63349
2026-05-26T21:37:37.020558+00:00 az900-vm1 sshd[11978]: pam_unix(sshd:auth): check pass; user unknown
2026-05-26T21:37:37.020681+00:00 az900-vm1 sshd[11978]: pam_unix(sshd:auth): authentication failure; logname= uid=0 euid=0 tty=ssh ru
ser= rhost=178.27.90.142
2026-05-26T21:37:39.659832+00:00 az900-vm1 sshd[11978]: Failed password for invalid user ubuntu from 178.27.90.142 port 63349 ssh2
2026-05-26T21:37:40.009074+00:00 az900-vm1 sudo: azureuser : TTY=pts/0 ; PWD=/home/azureuser ; USER=root ; COMMAND=/usr/bin/tail -f /
var/log/auth.log
2026-05-26T21:37:40.016840+00:00 az900-vm1 sudo: pam_unix(sudo:session): session opened for user root(uid=0) by azureuser(uid=1000)
2026-05-26T21:37:40.659949+00:00 az900-vm1 sshd[11978]: Received disconnect from 178.27.90.142 port 63349:11: Bye Bye [preauth]
2026-05-26T21:37:40.661160+00:00 az900-vm1 sshd[11978]: Disconnected from invalid user ubuntu 178.27.90.142 port 63349 [preauth]
2026-05-26T21:38:08.781147+00:00 az900-vm1 sshd[11991]: pam_unix(sshd:auth): authentication failure; logname= uid=0 euid=0 tty=ssh ru
ser= rhost=59.27.249.238 user=root
2026-05-26T21:38:10.677601+00:00 az900-vm1 sshd[11991]: Failed password for root from 59.27.249.238 port 58878 ssh2
2026-05-26T21:38:11.022062+00:00 az900-vm1 sshd[11993]: pam_unix(sshd:auth): authentication failure; logname= uid=0 euid=0 tty=ssh ru
ser= rhost=136.52.45.147 user=nobody
2026-05-26T21:38:13.329792+00:00 az900-vm1 sshd[11993]: Failed password for nobody from 136.52.45.147 port 62681 ssh2
2026-05-26T21:38:15.026222+00:00 az900-vm1 sshd[11991]: Failed password for root from 59.27.249.238 port 58878 ssh2
2026-05-26T21:38:18.559497+00:00 az900-vm1 sshd[11993]: Failed password for nobody from 136.52.45.147 port 62681 ssh2
2026-05-26T21:38:18.922214+00:00 az900-vm1 sshd[11991]: Failed password for root from 59.27.249.238 port 58878 ssh2
```

I connected the logs to Microsoft Sentinel
so I could see everything in one place.

Microsoft Sentinel | Logs ...

Selected workspace: 'law-sentinel-lab-001'

Run | Time range : Set in query Show : 1000 results KQL mode

```
1 Syslog
2 | where TimeGenerated > ago(1h)
3 | where Facility in ("auth", "authpriv")
4 | where SyslogMessage has "sshd"
5 | project TimeGenerated, HostName, Facility, SeverityLevel, SyslogMessage
6 | order by TimeGenerated desc
```

Results Chart Add bookmark

TimeGenerated [UTC]	HostName	Facility	SeverityLevel	SyslogMessage
5/26/2026, 9:56:49.442 PM	az900-vm1	authpriv	notice	PAM service(sshd)
5/26/2026, 9:56:32.419 PM	az900-vm1	authpriv	notice	pam_unix(sshd:au
5/26/2026, 9:55:16.259 PM	az900-vm1	authpriv	notice	pam_unix(sshd:au
5/26/2026, 9:55:16.259 PM	az900-vm1	authpriv	notice	pam_unix(sshd:au
5/26/2026, 9:54:14.853 PM	az900-vm1	authpriv	notice	pam_unix(sshd:au
5/26/2026, 9:54:14.852 PM	az900-vm1	authpriv	notice	pam_unix(sshd:au

This one made me really think.
One IP had tried 158 times.
The server had only been online
for a few hours. Most were Bots.

Microsoft Sentinel | Logs ...

ected workspace: 'law-sentinel-lab-001'

New Query 2* ... x New Query 1* +

Observability agent New Save Share ... Queries

Run | v Time range : Set in query Show : 1000 results KQL mode

Query editor

Results Chart Add bookmark

☐	SourceIP	Attempts	UsernameSeen	FirstSeen [UTC]	LastSeen [UTC]
☐	> 59.27.249.238	158	["root","admin","oracle","usuario","test","user","ftpuser","test1","test2","ubuntu","pi",...	5/26/2026, 9:38:10.677 PM	5/26/2026, 9:53:40.624 PM
☐	> 138.113.28.120	17	["root","bot","gitlab-runner","riad","novinhost","systemd","wg","ubuntu","www","ram...	5/26/2026, 9:59:18.089 PM	5/26/2026, 10:36:16.811 PM
☐	> 203.161.47.246	17	["bin","michel","homepage","root","devops","ubuntu","maman","wy","testa","redmin...	5/26/2026, 10:02:45.010 PM	5/26/2026, 10:29:52.804 PM
☐	> 95.46.211.142	17	["devops","bin","wy","administrador","testa","ubuntu","root","michel","homepage","s...	5/26/2026, 10:02:53.449 PM	5/26/2026, 10:30:05.648 PM
☐	> 34.123.134.194	17	["maman","devops","michel","root","wy","bin","systemd","ubuntu","homepage","test...	5/26/2026, 9:55:17.945 PM	5/26/2026, 10:32:19.250 PM
☐	> 178.27.90.142	16	["root","ts1","nigger","bruno","ubuntu","supermaint","postgres","systemd","api","dev...	5/26/2026, 9:30:57.292 PM	5/26/2026, 9:50:59.284 PM
☐	> 35.188.112.111	16	["steven","root","game","test1","db2","ubuntu","user","systemd","sammy","socks","te...	5/26/2026, 9:30:40.361 PM	5/26/2026, 9:51:22.316 PM
☐	> 45.148.10.152	14	["root"]	5/26/2026, 9:39:18.406 PM	5/26/2026, 10:13:24.215 PM
☐	> 152.32.205.153	12	["user","root","systemd","stud1","web","ociispth","ubuntu"]	5/26/2026, 10:03:08.654 PM	5/26/2026, 10:23:13.482 PM

I wrote the detection in KQL —
if any IP fails 10+ times in 1 hour,
flag it.

Microsoft Sentinel | Logs

Selected workspace: 'law-sentinel-lab-001'

New Query 1* ... x + Observability agent New Save v Share v ... | Quer

Run | v

Time range : Set in query

Show : 1000 results

KQL mode

```
1 let LookbackWindow = 1h;
2 let FailureThreshold = 10;
3 Syslog
4 | where TimeGenerated > ago(LookbackWindow)
5 | where Facility in ("auth", "authpriv")
6 | where SyslogMessage has "Failed password"
7 | extend SourceIP = extract(@"from (\S+) port", 1, SyslogMessage)
8 | extend Username = extract(@"for (?:invalid user )?(\\S+) from", 1, SyslogMessage)
9 | where isnotempty(SourceIP)
10 | summarize
11     FailureCount = count(),
```

Made it an automatic rule. Tagged with MITRE T1110
— the industry code for brute force attacks.

 SSH Brute Force — Linux Authentication

Severity	 Custom	 Enabled
■ ■ ■ Medium	Content Source	Status

Info Insights

7b14ff0d-d777-4c44-91fe-1c6389dade7f 

Description

Detects 10 or more SSH failed logins from a single source IP within 1 hour against Linux hosts ingested via Syslog AMA.

MITRE ATT&CK

✓  Credential Access (2)

✓ T1110 - Brute Force (1)

T1110.001 - Password Guessing

Rule query

```
let LookbackWindow = 1h;
let FailureThreshold = 10;
let TrustedIPs = dynamic(["136.52.45.147"]);
Syslog
```

Sentinel built an incident graph.

13 attacker IPs hitting my one server.

ID 258: SSH Brute Force — Linux Authentication

[Manage incident](#) [Tasks](#) [...](#)

■ ■ ■ Medium | ● Active | 👤 Unassigned | ✎ Unclassified | ⌚ Last update time: May 26, 2026 3:34 PM

Attack story Alerts (1) Activities (1) Assets (1) Investigations (0) Evidence and Response (13) Summary

Filters: Severity: Any ✕ Status: Any ✕ Service sources: Any ✕ [Add filter](#)

Detection & Categories

Active alerts

1/1

Categories

1

First activity

May 26, 2026 2:29:20 PM

Last activity

May 26, 2026 3:29:20 PM

Creation time

May 26, 2026 3:34:48 PM

Alerts

● May 26, 2026 2:29 PM ● New
SSH Brute Force — Linux Authentication
🖥 az900-vm1

[View incident details](#)

Incident graph

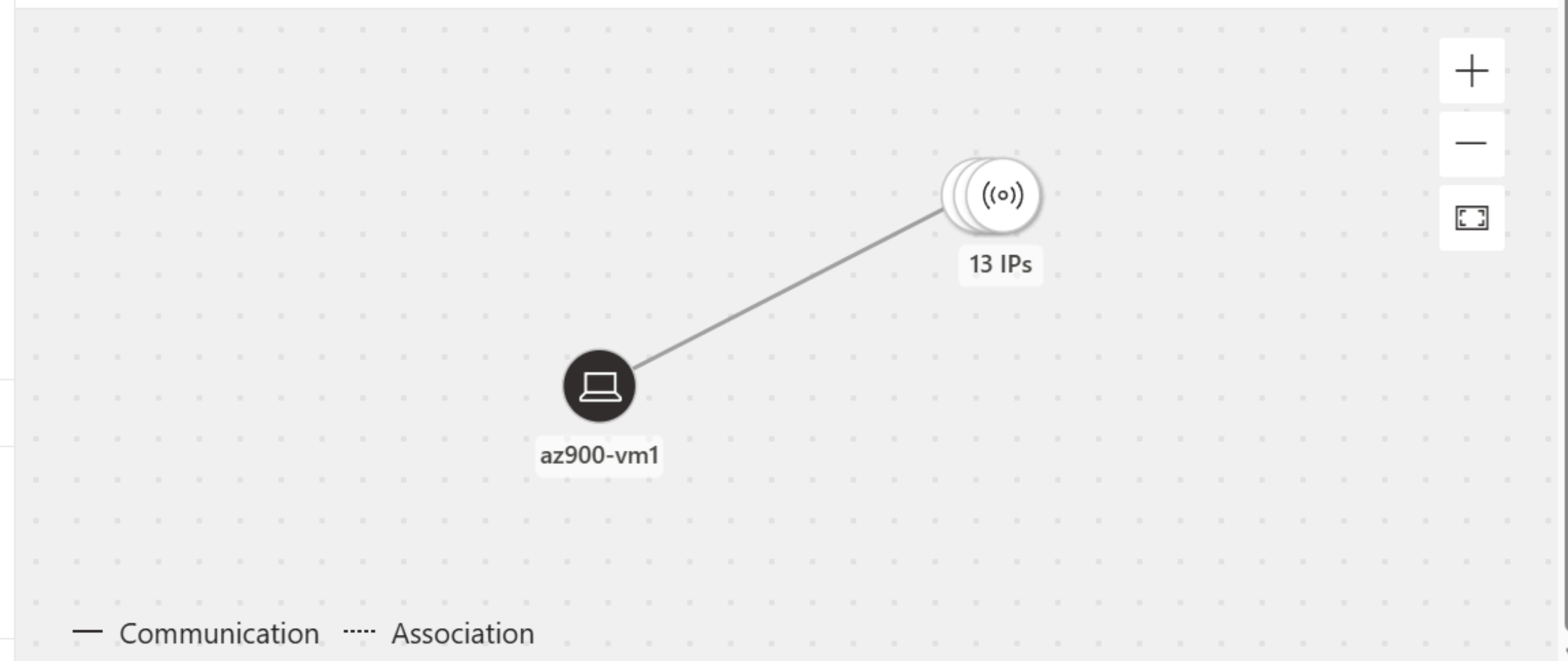
Layout ▾



Group similar nodes



Entity types ▾



Then I checked if any of them
actually got in.

Only one login worked — and it was mine.

Microsoft Sentinel | Logs ...

ed workspace: 'law-sentinel-lab-001'

New Query 1* ... × +

Observability agent New Save Share

Run | Time range : Set in query Show : 1000 results

```
1 Syslog
2 | where TimeGenerated > ago(2h)
3 | where SyslogMessage has_any ("Accepted password", "Accepted publickey")
4 | project TimeGenerated, HostName, SyslogMessage
5 | order by TimeGenerated desc
```

Results Chart | Add bookmark

☐	TimeGenerated [UTC]	HostName	SyslogMessage
☐	> 5/26/2026, 9:37:22.818 PM	az900-vm1	Accepted password for azureuser from 136.52.45.147 port 62675 ssh2

Real attacks. Real data. Real detection.

13 attacker IPs

158 attempts from one source

Zero successful logins

What this taught me:

-I didn't have to simulate attacks.

The internet did it for me.

- Failures are noise.

A success from the same IP

is the real alert.

**- Detecting attacks without stopping them
just means more alerts tomorrow.**